CSA6303- Threat Intelligence and Network Security

AT1: Application-Based Questions

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

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Total: 50

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Code of Conduct:

I A.Abhinava Reddy,192465054(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

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Cybersecurity Assignment Answers

Q1. CIA Triad — Scenario Identification (5 marks)

Violated Principle: Integrity
Justification: Integrity ensures that data is accurate, complete, and unmodified by unauthorized entities. Since the attacker gained database write access and silently modified the product prices, the data was tampered with, directly violating the Integrity principle. Availability was not impacted (as the site stayed fully online with no downtime), and Confidentiality, while potentially breached during the initial intrusion, is not the primary principle violated in the context of the unauthorized data alteration.

Q2. Risk Quantification — Annualized Loss Expectancy (10 marks)

Given Variables:
• Asset Value (AV) = $50,000
• Exposure Factor (EF) = 40% (0.40)
• Annualized Rate of Occurrence (ARO) = 2

Step 1: Calculate Single Loss Expectancy (SLE)
Formula: SLE = AV × EF
Calculation: $50,000 × 0.40 = $20,000

Step 2: Calculate Annualized Loss Expectancy (ALE)
Formula: ALE = SLE × ARO
Calculation: $20,000 × 2 = $40,000

Final Answers:
SLE = $20,000 | ALE = $40,000

Q3. Brute-Force Attack Time Estimation (10 marks)

Variables:
• Character Set = 62 characters (a-z, A-Z, 0-9)
• Password Length = 8 characters
• Cracking Speed = 1,000,000,000 (1 billion) passwords/second

Keyspace Calculation:
Formula: Keyspace = (Character Set) ^ (Length)
Keyspace = 62^8 = 218,340,105,584,896 possible combinations

Time Estimation (Worst-Case):
Formula: Time (seconds) = Keyspace / Cracking Speed
Time = 218,340,105,584,896 / 1,000,000,000 = 218,340.105 seconds
Time (hours) = 218,340.105 / 3600 = 60.65 hours

Final Answer:
The keyspace is 218,340,105,584,896, and the worst-case time to exhaust it is approximately 60.65 hours.

Q4. Risk Prioritization Matrix (10 marks)

Formula: Risk Score = Likelihood (L) × Impact (I)

Calculated Risk Scores:
• (a) Ransomware: L4 × I5 = 20
• (b) Phishing: L5 × I3 = 15
• (c) Insider data theft: L2 × I5 = 10
• (d) DDoS: L3 × I3 = 9
• (e) Unpatched server exploit: L4 × I4 = 16

Prioritization Ranking (Highest to Lowest Priority):
1. Ransomware (Score: 20)
2. Unpatched server exploit (Score: 16)
3. Phishing (Score: 15)
4. Insider data theft (Score: 10)
5. DDoS (Score: 9)

Q5. Passive vs. Active Threat Classification (5 marks)

(a) Silently sniffing unencrypted traffic on a public Wi-Fi network:
• Classification: Passive Threat
• CIA Principle Primarily Threatened: Confidentiality (data is observed but not altered or blocked).

(b) An attacker performing ARP spoofing to redirect traffic:
• Classification: Active Threat
• CIA Principle Primarily Threatened: Integrity and Confidentiality (data flows are manipulated, and traffic can be altered/intercepted via Man-in-the-Middle).

(c) A nation-state actor performing long-term network reconnaissance without altering anything:
• Classification: Passive Threat
• CIA Principle Primarily Threatened: Confidentiality (mapping networks and gathering intelligence without disruption).

Q6. Threat Intelligence Lifecycle sequencing (10 marks)

Correct Order and Respective Phases:

1. [3] The request itself is logged as a requirement.

Phase: Direction / Requirements & Planning

2. [4] OSINT and threat feeds are pulled for the group's known TTPs.
   Phase: Collection

3. [1] Analysts correlate collected IOCs with internal logs.
   Phase: Processing & Analysis

4. [2] A report is sent to the CISO.
   Phase: Dissemination

5. [5] The CISO's follow-up questions shape the next collection cycle.
   Phase: Feedback